PhenoBench: Benchmarking Scientific Agents for Image-Guided Perturbational Discovery

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Abstract

Current benchmarks for perturbational biology largely evaluate prediction: given a perturbation, forecast a cellular response or assign a phenotype to a predefined label. Yet in many experiments, the measured response is only the starting point of scientific analysis. We introduce PhenoBench, a benchmark for evaluating scientific agents on image-guided perturbational discovery. PhenoBench asks agents to reason from experimentally observed cellular phenotypes toward verifiable scientific conclusions, including functional relationships, mechanisms, liabilities, and candidates for follow-up. Its evaluations are grounded in real-world perturbation studies, preserve complementary molecular or functional evidence when required by the original scientific question, and do not prescribe a reference analysis pipeline. Instead, tasks are calibrated so that defensible analytical alternatives reach compatible endpoints while predefined shortcut and failure modes remain distinguishable; when necessary, controlled semi-synthetic variants are constructed from measurements of the same study to improve endpoint identifiability. In a pilot benchmark of 12 tasks spanning phenotype characterization, biological interpretation, and phenotype translation, we evaluate four frontier LLMs under a common agent harness with three independent runs per task. Overall pass rates range from 47.2% to 69.4%, with substantial variation across scientific objectives and no model achieving consistently reliable performance. These results suggest that frontier models can solve a meaningful fraction of open-ended perturbational analyses, while substantial challenges remain in turning experimental phenotypes into defensible scientific conclusions.

1 Introduction

A central goal of virtual-cell research is to predict how cells respond to genetic, chemical, or environmental perturbation, and recent benchmarks evaluate exactly this: forecasting transcriptional responses to unseen perturbations, or ranking genes expected to produce a desired assay phenotype [3, 20]. But a measured perturbation response is not the end of an experiment; it is the beginning of an analysis. Choosing controls, establishing what is reproducible, separating a real effect from a plate artifact, calibrating a claim to the evidence that supports it - this is the work that turns a screen into a conclusion, and prediction benchmarks do not measure it. The gap is widest for cellular morphology, where high-content assays such as Cell Painting are typically the initiating observation in phenotype-driven discovery rather than its output: a screen produces profiles, and the entire scientific question is what may be concluded from them - a mechanism, a target, a liability, or nothing at all. Computational work aimed at this setting has nonetheless kept the predictive framing [5]. AssayBench [7] formulates 1,920 public CRISPR screens as ranking genes from an assay description rather than analyzing the measurements those screens produced; CP-Agent [24] works directly with Cell Painting images but evaluates mechanism-of-action discrimination and the generation of mechanistic reports. Both ask what a perturbation will do, or which label a phenotype belongs to. Neither asks what a scientist could conclude from the experiment.

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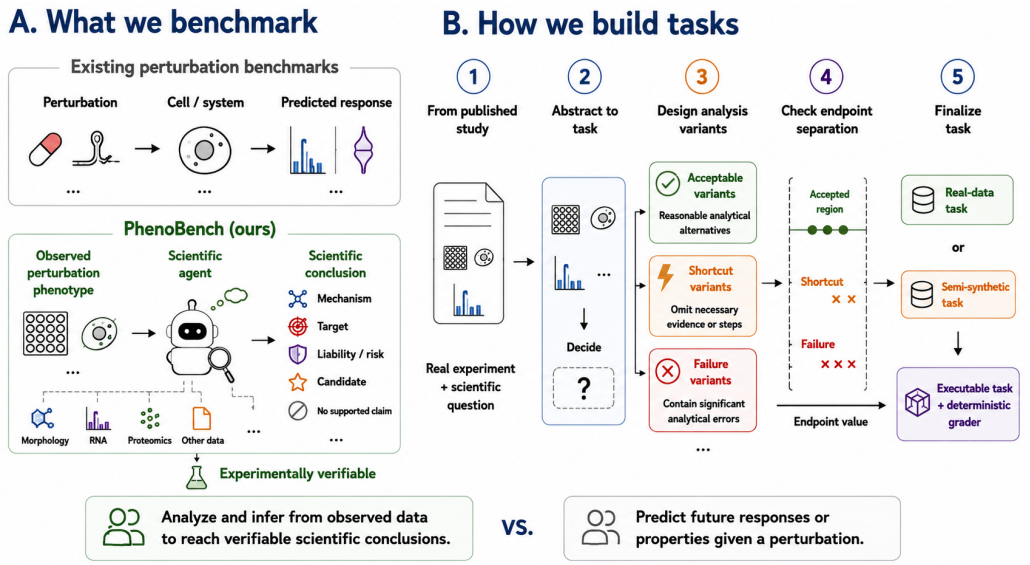


Figure 1: Overview of PhenoBench. (A) PhenoBench evaluates scientific reasoning from observed perturbational phenotypes rather than response prediction. (B) Tasks are grounded in published studies and calibrated for verifiable grading using real or controlled semi-synthetic data.

Whether an AI system can do that work—analyze a perturbation response rather than predict one—is a question benchmark design has only recently begun to ask. Evaluation of LLMs in biology has moved from knowledge-based question answering and prescribed tool-use workflows toward open-ended analysis, in which an agent receives staged data and a scientific question, chooses an analysis strategy, responds to intermediate results, and returns a verifiable conclusion. GeneBench-Pro [15], VariantBench [2], and scBench-Long [8] exemplify this shift. These and related process-level benchmarks [17] have focused primarily on sequence, variant, and tabular omics analyses. Perturbational phenotyping presents a complementary setting: the agent must reason from an experimentally observed cellular state, where morphology is often only the first piece of evidence and the scientific endpoint may require connecting that observation to mechanism, target, functional consequence, or follow-up intervention. This setting is particularly well suited to evaluating scientific agents because the underlying studies often contain not only rich phenotypic measurements, but also experimental controls, complementary assays, and downstream validation that can anchor the resulting scientific conclusion.

We introduce **PhenoBench**, as shown in Figure 1, a benchmark for scientific agents on image-guided perturbational discovery. PhenoBench begins where perturbation-response prediction ends: agents receive experimentally observed cellular phenotypes and must determine what can be concluded from them, including mechanisms, targets, liabilities, functional relationships, or candidates for follow-up. Its evaluations are study-grounded, originating from the scientific questions, experimental designs, and findings of published perturbation studies rather than from isolated question templates or predefined skill compositions. Cellular imaging serves as the initiating experimental observation, while complementary molecular and functional measurements may provide additional evidence when required by the scientific question. To make these open-ended analyses verifiable, PhenoBench evaluates agents against scientific endpoints grounded in the underlying experiments rather than against a prescribed reference pipeline. Tasks are calibrated so that defensible analytical alternatives remain within an accepted endpoint region while documented shortcut or incorrect analyses are separated from it; when the original measurements do not provide sufficient separation, controlled semi-synthetic variants are constructed from the same study data.

In a pilot evaluation comprising 12 tasks, we benchmark four frontier LLMs under a common agent harness with three independent runs per task. Overall pass rates range from 47.2% to 69.4%, with substantial variation across scientific analysis categories and no model approaching consistent success across the benchmark. These results suggest that current frontier models can complete a meaningful

Table 1: Scope of PhenoBench. Tasks are characterized along three complementary dimensions: their scientific objective, available evidence, and perturbational context.

• **Scientific objective** What scientific decision must the agent make?	◇ **Evidence** What experimental evidence is available?	○ **Perturbational context** What biological setting generates the phenotype?
Characterization & quality Establish and validate a perturbation-associated phenotype.	**Morphology** Image-derived cellular phenotypes.	**Genetic perturbation** Gene overexpression, knockout, or knockdown.
Biological interpretation Determine relationships and biological implications of the phenotype.	**Molecular evidence** Transcriptomic, genomic, or other molecular measurements.	**Chemical perturbation** Compound treatment and dose response.
Translation & prioritization Use phenotypic evidence to guide downstream discovery decisions.	**Functional evidence** Phenotypic assays, viability, or related experimental readouts.	**Genetic variation** Disease-associated or engineered variants.
	Multiple experiments Independent imaging or perturbation experiments.	**Cross-perturbation** Comparisons across genetic and chemical systems.

fraction of open-ended perturbational analyses, while substantial gaps remain in reliable scientific reasoning over experimental data.

2 PhenoBench Design

2.1 Study-Grounded Evaluation Discovery

PhenoBench evaluations are derived from published perturbational studies rather than assembled from isolated question templates or predefined skill combinations. We identify studies in which cellular imaging plays a substantive role in investigating genetic, chemical, or environmental perturbations. Candidate studies are selected according to four criteria: (1) cellular morphology provides an initiating or decision-relevant experimental observation; (2) the study addresses a concrete downstream scientific question, such as identifying a mechanism, target, functional relationship, liability, or candidate intervention; (3) the measurements and experimental metadata required to revisit that question are publicly available or reproducibly recoverable; and (4) the study provides a scientific outcome that can be independently supported by the deposited data or downstream experimental validation.

We exclude studies whose central conclusion cannot be reconstructed from available measurements, depends primarily on qualitative interpretation, or would be directly revealed by the information provided to the agent. Selection is based on the scientific question and experimental evidence of the study rather than on whether its published computational workflow can be reproduced exactly.

2.2 From Published Studies to Agent Tasks

For each selected study, we identify the scientific decision connecting the perturbation experiment to its downstream finding and abstract this decision into a gradable endpoint. The benchmark preserves the original scientific objective while removing instructions that prescribe a particular analysis procedure. Agents are therefore given the experimental context and measurements needed to answer the scientific question, but are free to choose how those data are analyzed.

Each task consists of an initiating phenotypic observation, agent-visible experimental evidence, and a scientific endpoint. Although cellular imaging provides the starting point across PhenoBench, the resulting evaluations span different scientific objectives, evidence compositions, and perturbational contexts, as summarized in Table 1. Many source studies require complementary evidence from transcriptomic, proteomic, metabolomic, genetic, or functional assays; these modalities are preserved

when they are necessary to interpret the phenotype. PhenoBench is therefore image-centered rather than image-only.

Information that directly reveals the reported conclusion, such as downstream validation labels, author-generated candidate rankings, or derived outputs that effectively prescribe the published analysis, is withheld. Each task terminates in a structured scientific endpoint, including numerical, categorical, ranked, or set-valued outputs that can be evaluated reproducibly. Table 2 illustrates three representative evaluations [14, 18, 19] spanning phenotype quality assessment, biological interpretation, and downstream discovery prioritization.

2.3 Analysis Variants and Endpoint Validation

The published analysis provides one successful solution to each task but does not define the only acceptable one. For each candidate evaluation, we therefore construct a small set of representative analysis variants. These include acceptable variants reflecting reasonable analytical alternatives and plausible failure variants representing shortcuts, omitted necessary steps, or scientifically consequential analytical errors.

Variants are initialized from the analysis described in the source publication and, when available, its released code, and are subsequently refined through manual analysis and review by domain experts. We execute each variant on the staged benchmark data and compare the resulting endpoints. A task is directly gradable when acceptable variants support compatible scientific conclusions while predefined failure variants produce distinguishable endpoints. Thus, PhenoBench aims to identify a verifiable scientific conclusion without requiring a unique analytical workflow.

2.4 Task Calibration and Grading

When the original study data provide sufficient separation between acceptable and failure variants, the original measurements are used directly. Otherwise, we construct a controlled semi-synthetic task from measurements originating from the same study. Such modifications are task-specific and alter only the elements required to improve endpoint identifiability while retaining the biological setting and empirical characteristics of the source experiment.

The outputs of acceptable analysis variants define the task-specific acceptance criterion, while failure variants are used to verify that the criterion does not admit predefined incorrect solutions. Numerical tasks are evaluated using calibrated acceptance ranges, while categorical, ranked, and set-valued tasks use corresponding deterministic grading rules.

Each evaluation is delivered as a scientific prompt together with staged experimental files in an isolated analysis environment. Details of the evaluated agent systems, available tools, execution budgets, repeated trials, and aggregate metrics are described separately in the experimental setup.

3 Results

3.1 Experimental Setup

We evaluate PhenoBench using a fixed agent harness to isolate differences in model capability from differences in agent scaffolding. All experiments are executed through the Harbor [11] evaluation framework using Mini-SWE-Agent [23] as the common agent environment. Each evaluation is instantiated in an isolated container containing the task prompt, staged experimental files, and the software required for local data analysis. The same harness configuration and task environment are used across all models.

We evaluate four frontier models: GPT-5.6 Sol [16], Claude Opus 4.8 [1], Grok 4.6 [21], and Gemini 3.6 Flash [10]. Models are accessed through Microsoft Azure and Google Cloud Platform, depending on provider availability. Reasoning is enabled at the highest available effort setting for each model. Apart from provider-specific API requirements, no model-specific prompts, tools, or agent modifications are introduced. Following repeated-attempt evaluation protocols used in recent scientific-agent benchmarks, each model is evaluated independently three times on every PhenoBench task. With 12 evaluations and four models, this yields 144 agent trajectories in total. Each trajectory is graded independently using the frozen task-specific verifier described in Section 2.4. Unless otherwise

Table 2: Representative PhenoBench tasks. The examples span distinct scientific objectives while sharing a common structure: analysis begins from an observed perturbational phenotype and terminates in a verifiable scientific endpoint.

Representative task	Scientific question	Agent-visible evidence	Verifiable endpoint
Chromosome-arm bias correction	Is an apparent gene–gene phenotypic relationship driven by CRISPR chromosome-arm proximity bias, and can that effect be removed without erasing biological signal?	Genome-wide CRISPR Cell Painting profiles together with genomic and experimental-context annotations.	A corrected gene-similarity representation with reduced chromosome-arm bias while retaining recoverable biological relationships.
Cross-modal functional discovery	Which perturbations exhibit reproducible and biologically coherent relationships across cellular morphology and gene expression?	Cell Painting profiles together with matched L1000 expression profiles and perturbation metadata.	Signed gene relationships and functional modules supported across the available modalities.
Image-guided compound prioritization	Which compounds phenocopy or oppose the cellular morphology induced by a target genetic perturbation?	Independent genetic-perturbation and compound-treatment Cell Painting profiles with experimental metadata.	Ranked sets of compounds that phenocopy or oppose the target perturbation phenotype.

stated, we report endpoint pass rate across all attempts as the primary metric and additionally examine task-level consistency across the three repeated runs.

3.2 Overall Performance and Efficiency

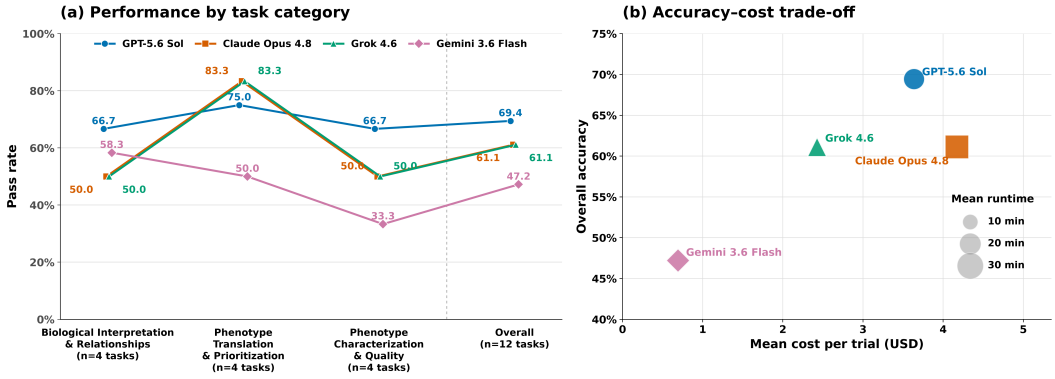


Figure 2: Performance and efficiency on PhenoBench. **(a)** Pass rates across three scientific-analysis categories and overall. Each of the 12 tasks is evaluated with three independent runs per model. **(b)** Overall pass rate versus mean API cost per trial; marker size denotes mean runtime.

Across the 12 PhenoBench evaluations, all four frontier models solve a substantial fraction of the tasks, but none achieves consistently reliable performance (Figure 2). GPT-5.6 Sol obtains the highest overall pass rate at 69.4%, followed by Claude Opus 4.8 and Grok 4.6 at 61.1%, while Gemini 3.6 Flash achieves 47.2%. Because each task is evaluated over three independent runs, these results reflect 36 trajectories per model rather than a single attempt on each evaluation.

Performance varies substantially across scientific objectives. Phenotype Translation & Prioritization is the strongest category for Claude Opus 4.8 and Grok 4.6, both reaching 83.3%, and also for GPT-5.6 Sol at 75.0%. In contrast, performance on Phenotype Characterization & Quality falls to 50.0% for Claude and Grok and 33.3% for Gemini. GPT-5.6 Sol is comparatively stable across categories, ranging from 66.7% to 75.0%. These differences suggest that aggregate performance alone masks substantial variation in the types of scientific analyses that current models can complete reliably.

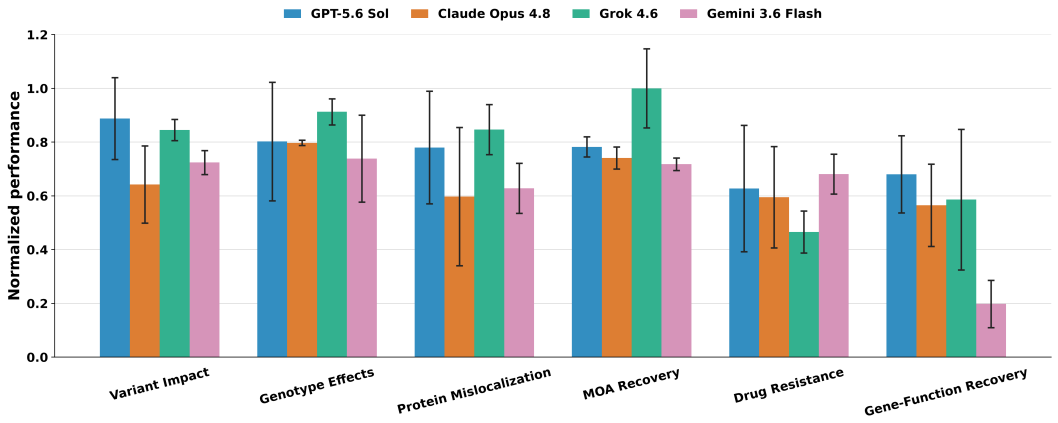


Figure 3: **Continuous performance on representative PhenoBench evaluations.** Scores are normalized by the corresponding reference-analysis value, with 1.0 indicating reference-level performance. Bars show the mean across all three independent runs and error bars denote one standard deviation.

The models also occupy distinct performance–cost trade-off. GPT-5.6 Sol achieves the highest accuracy at a moderate-to-high mean cost per trial, whereas Grok 4.6 reaches the same overall pass rate as Claude Opus 4.8 at a substantially lower mean cost. Gemini 3.6 Flash is the least expensive model evaluated but also has the lowest pass rate. Mean runtime similarly varies across models, indicating that higher scientific-agent performance does not correspond to a single uniform computational profile.

3.3 Partial Progress and Representative Failure Modes

Binary pass/fail outcomes do not capture variation in endpoint quality across agent trajectories. We therefore examine six tasks [4, 6, 9, 12, 13, 22] whose endpoints admit a continuous performance measure and normalize each score by the corresponding reference-analysis value, such that a value of 1.0 represents reference-level performance. As shown in Figure 3, frontier models often recover substantial portions of the target signal even when they do not satisfy the task-specific acceptance criterion. Across most evaluations, normalized performance lies between approximately 0.6 and 0.9, although both model and task effects are pronounced. Grok 4.6 approaches reference performance on MoA recovery and performs strongly on genotype effects and protein mislocalization, whereas GPT-5.6 Sol is comparatively strong on variant impact. In contrast, gene-function recovery is particularly challenging for Gemini 3.6 Flash. Variation across repeated runs further indicates that endpoint quality is not always reproduced consistently. These continuous outcomes complement the binary benchmark scores by revealing variation in endpoint quality across all trajectories. To examine why some analytically substantial trajectories nevertheless fail verification, we next inspect two representative failure cases.

Representative trajectories reveal that failure often occurs after substantial analytical progress rather than at the level of basic execution (Figure 4). In the gene-function task, the agent recognized strong feature redundancy and explicitly evaluated whitening, but selected the unwhitened representation because it improved split-replicate reproducibility. The resulting profiles were reproducible yet failed to preserve the biological neighborhood structure required by the endpoint. In the drug-resistance task, the agent similarly recognized the small-sample, high-dimensional setting and constructed a sparse signature with perfect leave-one-out discrimination, but did not test whether the selected features were specific to resistance rather than to individual clones; the signature consequently failed to transfer to held-out clones. Together, these cases illustrate a recurring gap between locally reasonable analytical decisions and the scientific criterion that must ultimately be satisfied.

4 Conclusion

We introduced PhenoBench, a pilot benchmark for evaluating scientific agents on image-guided perturbational discovery. Rather than testing perturbation-response prediction alone, PhenoBench

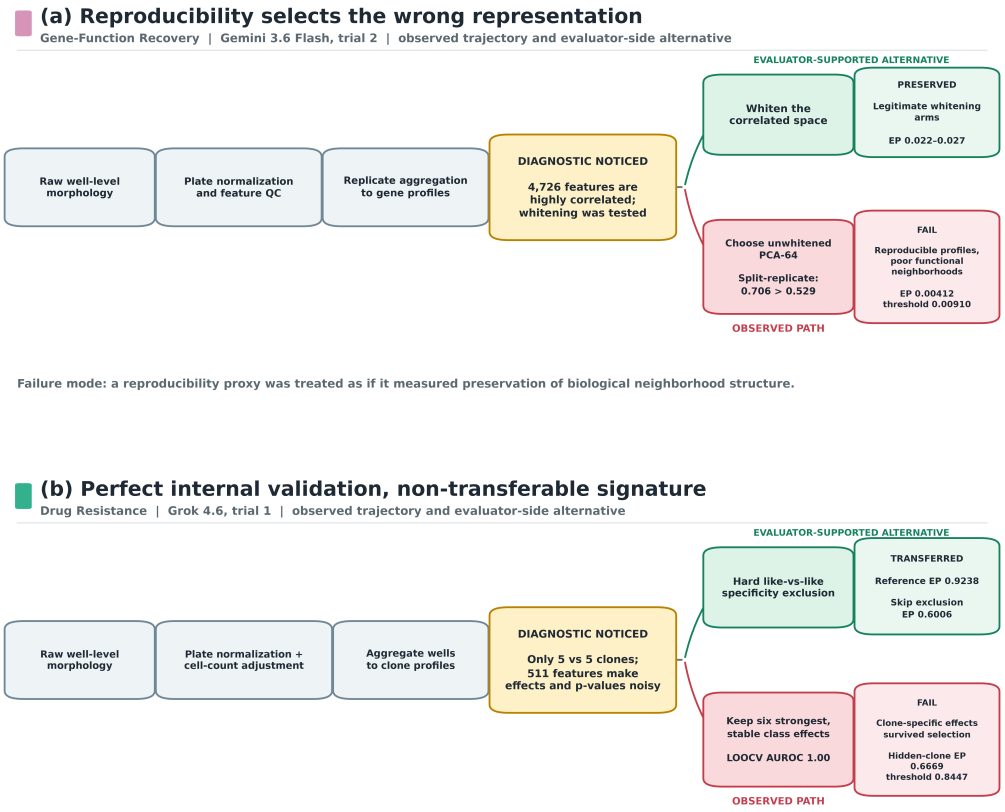


Figure 4: Representative failure trajectories in PhenoBench. **(a)** A model recognizes feature redundancy and tests whitening, but selects an unwhitened representation using replicate reproducibility, which does not preserve the functional neighborhood structure required by the task. **(b)** A model achieves perfect leave-one-out discrimination on the observed drug-resistance clones, but omits a specificity control required for transfer to unseen clones. Green branches show evaluator-supported alternatives; red branches show the observed model trajectories.

asks agents to analyze experimentally observed cellular phenotypes and derive verifiable scientific conclusions. Its tasks are grounded in real-world perturbation studies, preserve multimodal evidence when required by the original scientific question, and are calibrated to allow defensible analytical alternatives while rejecting consequential analytical failures.

Across our pilot evaluation of frontier LLMs under a common agent harness, overall task pass rates were approximately 60%, indicating that current models can solve a substantial fraction of these open-ended scientific analyses but remain far from consistently reliable. As a pilot study, PhenoBench is currently limited to 12 evaluations, several of which require controlled semi-synthetic construction to obtain discriminative and reproducible endpoints, and our model comparison is conducted under a single common harness. These limitations constrain the breadth of conclusions that can be drawn about scientific-agent performance, but also define clear directions for extending the benchmark.

Future work will focus on the next phase of PhenoBench, expanding the benchmark toward approximately 100 evaluations spanning a broader range of perturbational studies and evaluating a wider set of models and agent harnesses. This expansion will enable a more systematic characterization of how task domain, model capability, and scientific scaffolding jointly determine performance.

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